

Industrial Internship Report

Music Player with Playlist using Java, JDBC and MySQL

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Executive Summary

This report presents the development of a Music Player with Playlist Management System during the industrial internship. The project was developed using Java, JDBC and MySQL. The application allows users to manage songs, create playlists, add or remove songs from playlists and simulate playback operations such as Play, Pause, Stop and Skip. The project enhanced my understanding of object-oriented programming, database connectivity and software development.

1. Preface

The internship provided valuable industrial exposure and practical experience in Java application development. During the six-week internship I designed and implemented a Music Player application with playlist management. The project required problem analysis, database design, coding, testing and documentation. I sincerely thank upskill Campus, The IoT Academy, UniConverge Technologies Pvt. Ltd. and my faculty mentors for their guidance.

2. Introduction

The Music Player with Playlist Management System is a Java application connected to a MySQL database through JDBC. It stores songs and playlists permanently and allows users to perform playlist operations. The application demonstrates the practical implementation of Java OOP concepts, JDBC API and SQL queries.

2.1 Objectives

- Gain practical knowledge of Java and JDBC.
- Understand database connectivity.
- Build a real-world application.
- Improve coding, debugging and testing skills.
- Learn GitHub project management.

3. Problem Statement

Many simple music player applications do not provide database-based playlist management. The objective of this project is to develop a Music Player that stores song information in MySQL, supports playlist creation, allows adding and removing songs from playlists and provides playback controls.

4. Existing and Proposed Solution

Existing applications often store playlist information temporarily or provide limited educational examples. The proposed solution uses Java with JDBC to connect to MySQL, making song and playlist information persistent. The application is modular, easy to maintain and demonstrates industry-standard database operations.

Code Submission (GitHub): _____

Report Submission (GitHub): _____

5. Proposed Design / Model

Modules:

- Database Connection Module
- Song Management Module
- Playlist Management Module
- Playback Control Module
- User Interface Module

High Level Flow:

User → Java Application → JDBC → MySQL Database

Features:

1. Display Songs
2. Create Playlist
3. Add Songs to Playlist
4. View Playlist
5. Remove Songs
6. Play, Pause, Stop and Skip Songs

6. Performance Testing

Test Cases:

- Display songs from database.
- Create a new playlist.
- Add and remove songs from playlist.
- Verify playback commands.

Result:

All core functions executed successfully and database operations produced correct results without data loss.

7. My Learnings

The internship strengthened my understanding of Java programming, OOP concepts, JDBC connectivity, SQL queries, database design, exception handling, modular programming, debugging and version control using GitHub. It also improved my confidence in developing complete software projects.

8. Future Work Scope

Future enhancements include a graphical user interface using JavaFX or Swing, real audio playback support, user authentication, search functionality, favourite songs, shuffle and repeat modes, cloud synchronization and mobile integration.